Liwen Zou, Ph.D.

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Bio: Liwen Zou is a postdoctoral fellow at **the Hong Kong Polytechnic University**, working with ***Prof. Defeng Sun***. Previously, he earned his Ph.D. at **Nanjing University**, supervised by ***Prof. Xiaoping Yang***. His research focuses on AI-driven healthcare, encompassing core medical imaging tasks including **disease diagnosis, lesion detection, image segmentation and spatial registration**, alongside the development of **medical foundation models**. His key contributions lie in AI-enabled precision diagnosis and assessment of **pancreatic cancer** via CT imaging, with relevant findings published in top journals including ***Medical Image Analysis*** and ***Artificial Intelligence in Medicine***. He has constructed a large-scale pancreatic cancer imaging dataset with high-quality annotations, covering **over 29,000 patients** from **17 hospitals** across China, and established long-term partnerships with these medical centers to advance high-impact clinical research and translational medicine.

Education

2020-09 – 2025-12 **Ph.D. in Applied Mathematics, Nanjing University** Nanjing, China
Supervisor: ***Prof. Xiaoping Yang***
Thesis: ***Mechanism Modeling, Clinical Thinking and Deep Learning Fused Medical Image Segmentation Methods for Precision Diagnosis and Treatment of Pancreatic Cancer***

2016-09 – 2020-06 **B.S. in Information and Computing Science, South China University of Technology** Guangzhou, China
GPA: 3.80/4.00 **Ranking**: 2/35

Research Publications

Journal Articles

1. **L. Zou**, Y. Cao, Z. Nie, L. Mao, Y. Qiu, Z. Wang, Z. Cai, and X. Yang, "Segment like a doctor: Learning reliable clinical thinking and experience for pancreas and pancreatic cancer segmentation," **Medical Image Analysis**, vol. 102, p. 103 539, 2025, DOI GitHub.
2. **L. Zou**, Z. Cai, L. Mao, Z. Nie, Y. Qiu, and X. Yang, "Automated peripancreatic vessel segmentation and labeling based on iterative trunk growth and weakly supervised mechanism," **Artificial Intelligence in Medicine**, vol. 150, p. 102 825, 2024, DOI GitHub.
3. **L. Zou**, Z. Cai, Y. Qiu, L. Gui, L. Mao, and X. Yang, "Ctg-net: An efficient cascaded framework driven by terminal guidance mechanism for dilated pancreatic duct segmentation," **Physics in Medicine & Biology**, vol. 68, no. 21, p. 215 006, 2023, DOI.
4. **L. Zou**, X. Luo, D. Zeng, L. Ling, and L. C. Zhao, "Measuring the rogue wave pattern triggered from gaussian perturbations by deep learning," **Physical Review E**, vol. 105, no. 5, p. 054 202, 2022, DOI GitHub.
5. H. Liu, **L. Zou**, N. Xu, H. Shen, Y. Zhang, P. Wan, B. Wen, X. Zhang, Y. He, L. Gui, et al., "Deep learning radiomics based prediction of axillary lymph node metastasis in breast cancer," **NPJ Breast Cancer**, vol. 10, no. 1, p. 22, 2024, **(Co-first authors)** DOI GitHub.
6. Z. Cai, **L. Zou**, Q. Li, J. Chen, Y. Qiu, J. Ji, and L. Mao, "Deep learning to predict extrapancreatic perineural invasion at ct images," **Annals of Medicine**, vol. 57, no. 1, p. 2 568 116, 2025, **(Co-first authors)** DOI.

7. C. Ren, **L. Zou**, and L. Gui, "2am: Weakly supervised tumor segmentation in pathology via cam and sam synergy," **Electronics**, vol. 14, no. 15, p. 3109, 2025, **(Co-corresponding authors)** DOI.

8. J. Lin, **L. Zou**, Y. Gao, L. Mao, and Z. Nie, "Learning the anatomical topology consistency driven by wasserstein distance for weakly supervised 3d pancreas registration in multi-phase ct images," **Biomedical Physics & Engineering Express**, 2026, **(Co-corresponding authors)** DOI.

Conference Papers

1. Y. Zhu, **L. Zou**, L. Li, and P. Wen, "Selected partially labeled learning for abdominal organ and pan-cancer segmentation," pp. 209–221, **MICCAI Challenge on Fast and Low-Resource Semi-supervised Abdominal Organ Segmentation**, 2023, Springer, 2023. DOI GitHub.

2. Y. Zhu, **L. Zou**, P. Wen, Z. Nie, L. Gui, and X. Yang, "A novel pseudo label-based unsupervised multiple target domain adaptation framework for abdominal organ segmentation," pp. 164–177, **MICCAI Challenge on Fast and Low-Resource Semi-supervised Abdominal Organ Segmentation**, 2024, Springer, 2024. DOI.

Research Projects

2022-04 – 2023-04 **Accurate and Efficient AI Models for Pancreatic Cancer and Peripancreatic Vessel Segmentation on CT Images.** (Role: **Project Leader**; Funding: Postgraduate Research and Practice Innovation Program of Jiangsu Province, China, No. KYCX22_0082)

2020-12 – 2025-12 **Mathematical Theories and Algorithms for Precision Diagnosis and Treatment of Pancreatobiliary Malignant Tumors.** (Role: **Subproject Manager**; Funding: Ministry of Science and Technology of the People's Republic of China, No. 2020YFA0713802)

Additional Experience

Awards and Achievements

2024 **National Scholarship for Doctoral Students of China**

Outstanding Graduate Student Model of Nanjing University

2022 **Honorable Mention Award, MICCAI PARSE 2022: Pulmonary Artery Segmentation Challenge**

2019 **Grand Prize, Mathematics Competition of South China University of Technology**

2018 **Second Prize, National College Student Mathematics Competition of China**

Patents

2023 Yang, X., **Zou, L.,** Cai, Z., Qiu, Y., Gui, L., Mao, L. "A Dilated Pancreatic Duct Segmentation Method Based on Cascaded Terminal Guidance Mechanism." China Patent, No. 202310639583.1

Yang, X., **Zou, L.,** Cai, Z., Qiu, Y., Nie, Z., Mao, L. "A Peripancreatic Vessel Segmentation and Labeling Method Based on Iterative Trunk Growth and Weakly Supervised Learning." China Patent, No. 202310639579.5

Technical Demo

2026 AI-based diagnosis, segmentation and resectability assessment of pancreatic cancer from CT imaging. DEMO